

The primary differences between **LVDS** (Low-Voltage Differential Signaling) and **subLVDS** (sub-Low-Voltage Differential Signaling) lie in their electrical characteristics, voltage levels, and intended applications. Below is a concise comparison based on standard definitions and the context of Lattice CrossLink-NX FPGAs:

1. Voltage Levels (VCCIO):

- **LVDS:** Operates with a higher supply voltage, typically requiring a VCCIO of 2.5V or 3.3V in FPGA I/O banks (per the TIA/EIA-644-A standard). The common-mode voltage is around 1.2V, with a differential swing of approximately 350 mV.
- **subLVDS:** Designed for lower voltage operation, typically using a VCCIO of 1.8V. It has a lower common-mode voltage (around 0.9V) and a smaller differential swing (around 200–300 mV), making it more power-efficient for specific low-voltage applications.

2. Power Consumption:

- **LVDS:** Consumes more power due to higher voltage levels and larger differential swing.
- **subLVDS:** Lower power consumption due to the 1.8V VCCIO and reduced signal swing, suitable for power-sensitive designs.

3. Applications:

- **LVDS:** A widely used differential signaling standard for high-speed data transmission (up to 1.5 Gbps in CrossLink-NX), commonly found in displays, high-speed serial links, and general-purpose differential interfaces. It's robust for long-distance transmission due to its higher signal integrity.
- **subLVDS:** Primarily used in image sensor interfaces, such as those in cameras (e.g., Sony image sensors), where low power and 1.8V compatibility are critical. It's less common for general-purpose applications compared to LVDS.

4. Signal Characteristics:

- **LVDS:** Higher differential output voltage (VOD) of ~250–400 mV and a common-mode voltage of ~1.2V. It's designed for high noise immunity and longer trace lengths.
- **subLVDS:** Lower VOD (~150–300 mV) and common-mode voltage (~0.9V), optimized for shorter trace lengths and low-power sensor interfaces.

5. Compatibility with CrossLink-NX:

- **LVDS:** Supported on CrossLink-NX FPGA I/O banks (Bank 1 and Bank 2) with VCCIO typically set to 2.5V. It supports high-speed differential signaling up to 1.5 Gbps.
- **subLVDS:** Also supported on CrossLink-NX, but with VCCIO set to 1.8V, making it compatible with low-voltage interfaces like those used in some image sensors. It's explicitly mentioned in Lattice documentation for camera interface applications.

6. Interfacing with HyperRAM:

- As noted in prior responses, HyperRAM typically uses single-ended 1.8V LVCMOS signaling. **subLVDS** is more voltage-compatible (1.8V VCCIO) but still unsuitable due to its differential nature, while **LVDS** is incompatible due to both higher voltage (2.5V) and differential signaling requirements.

Summary:

- **LVDS:** Higher voltage (2.5V/3.3V), larger signal swing, higher power, used for general-purpose high-speed differential signaling.
- **subLVDS:** Lower voltage (1.8V), smaller signal swing, lower power, tailored for specific low-voltage applications like image sensors. For interfacing with a 1.8V HyperRAM, neither LVDS nor subLVDS is ideal due to HyperRAM's single-ended LVCMOS18 requirement. Use LVCMOS18 instead, as confirmed by successful implementations on CrossLink-NX at 1.8V.

For precise details, refer to the CrossLink-NX datasheet or Lattice Radiant software for supported I/O standards and electrical specifications.